

Synergy Network

Master Network, Ports, and Nginx Configuration Manual

This document is the single source of truth binding DNS, nginx, service ports, SDK configuration, and operational security rules for Synergy Network.

GLOBAL

1. Canonical DNS & Endpoint URLs

Purpose	URL
Global API Router	https://api.synergy-network.io
Global RPC Router	https://rpc.synergy-network.io
Global WS Router	wss://ws.synergy-network.io
Explorer Hub	https://explorer.synergy-network.io
Indexer Hub	https://indexer.synergy-network.io
Validator Portal	https://validators.synergy-network.io
Status Page	https://status.synergy-network.io
Static CDN	https://assets.synergy-network.io
Developer Docs	https://docs.synergy-network.io
Bootnode 1	bootnode1.synergy-network.io
Bootnode 2	bootnode2.synergy-network.io
Bootnode 3	bootnode3.synergy-network.io

1.1 Bootnode Roles & Classification

Bootnodes are classified as either TEMPORARY or CANONICAL.

TEMPORARY BOOTNODES

- May be replaced, migrated, or decommissioned without governance events
- Exist to facilitate network bootstrap, migrations, or emergency cutovers
- MUST NOT be treated as long-term trust anchors

CANONICAL BOOTNODES

- Intended for long-term stability
- Changes require explicit governance approval
- Serve as persistent discovery anchors across network lifecycles

Bootnode 1

bootnode1.synergy-network.io

Role: **TEMPORARY**

Bootnode 2

bootnode2.synergy-network.io

Role: **CANONICAL**

Bootnode 3

bootnode3.synergy-network.io

Role: **CANONICAL**

2. Global Router VHosts (Path-Based Routing)

2.1 Global RPC Router

```
server {
    listen 443 ssl http2;
    server_name rpc.synergy-network.io;
    include snippets/ssl.conf;

    location /devnet/ {
        proxy_pass http://devnet_core_rpc/;
    }

    location /healthz { return 200 "ok\n"; }
    location /readyz { return 200 "ready\n"; }
    location /version { return 200 "rpc-router-v1\n"; }
}
```

2.2 Global API Router

```
server {
    listen 443 ssl http2;
    server_name api.synergy-network.io;
    include snippets/ssl.conf;

    location /devnet/ {
        proxy_pass http://devnet_api/;
    }

    location /healthz { return 200 "ok\n"; }
    location /readyz { return 200 "ready\n"; }
    location /version { return 200 "api-router-v1\n"; }
}
```

2.3 Global WebSocket Router

```
server {
    listen 443 ssl http2;
    server_name ws.synergy-network.io;
    include snippets/ssl.conf;

    location /devnet/ {
        proxy_pass http://devnet_core_ws/;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

3. Security & Hard Rules

This section defines non-negotiable security and exposure rules for all Synergy Network infrastructure across all environments.

Violations of these rules constitute a misconfiguration and are considered out of spec.

Public Service Exposure

- All public-facing application services **MUST** be exposed exclusively over HTTPS (port 443).
- No other public-facing ports are permitted for HTTP, RPC, or WebSocket access.

P2P Network Isolation

- All P2P services (including bootnodes and validator nodes) **MUST** be DNS-addressed only.
- P2P ports **MUST NOT** be proxied, routed through nginx, CDN, or TLS termination layers.
- P2P traffic is strictly out-of-band from application-layer routing.
- Bootnodes **MUST NOT** be colocated with Global Routers, Environment Routers, API Gateways, or nginx instances.
- Bootnodes are discovery-only infrastructure and **MUST** remain operationally isolated from all application-layer routing.

Bootnode Redundancy Requirement

- All Synergy clients (nodes, validators, explorers, indexers, wallets, SDKs) **MUST** be configured with a minimum of TWO (2) bootnodes.
- Clients **MUST NOT** assume the availability, permanence, priority, or trust primacy of any single bootnode.
- Bootnodes are discovery mechanisms, not dependency endpoints.

Key Management & Sensitive Services

- Aegis KMS **MUST NEVER** be exposed publicly under any circumstances.
- Aegis KMS endpoints are private, mTLS-protected, and environment-scoped only.

Metrics & Observability

- Metrics endpoints **MUST** be bound to localhost only.
- Public or externally routable metrics exposure is prohibited.

Mainnet Write Protection

- Mainnet write operations **MUST** require explicit routing and configuration.
- Implicit or default write paths to Mainnet are prohibited.

Specification Authority

- If a service, port, hostname, or routing rule is not defined in this document, it is **NOT** considered official and **MUST NOT** be deployed.

4. Client Bootstrap Policy (Global)

Bootnodes exist for INITIAL PEER DISCOVERY ONLY.

All Synergy clients **MUST** adhere to the following:

1. Clients **MUST** attempt all configured bootnodes in parallel during startup.
2. Clients **MUST** randomize peer selection after discovery.
3. Clients **MUST** drop bootnode connections once a healthy peer threshold is reached.
4. Clients **MUST NOT** rely on bootnodes for sustained connectivity.
5. Clients **MAY** re-bootstrap only if peer count drops below threshold.
6. Clients **MUST** treat bootnode connections as transient.

7. Clients SHOULD impose conservative timeouts on bootnode connections and MUST NOT block startup, synchronization, or operation on sustained bootnode availability.

Failure to comply with this policy constitutes a client implementation bug.

5. Bootnode Network Surface Rules

Bootnodes are P2P-only infrastructure.

Bootnodes MUST NOT expose:

- *HTTP*
- *HTTPS*
- *RPC*
- *WebSockets*
- *nginx*
- *TLS termination*
- *Application-layer health endpoints*

Bootnodes MUST NOT be used for:

- *Chain state queries*
- *Block propagation guarantees*
- *Indexing*
- *Explorer backends*
- *Wallet balance resolution*
- *Any form of authoritative data sourcing*

Bootnode liveness is inferred solely through P2P reachability.

6. Change Control (Global)

This document is the authoritative and binding contract governing DNS records, nginx configuration, service exposure, and port allocation for the Synergy Network.

Change Authority

- This document is the single source of truth for all environments.
- All future environments (Devnet, Testnet, Mainnet-Beta, Mainnet) MUST extend this document explicitly.
- No environment may override, bypass, or reinterpret rules defined herein.

Override Prohibition

- Silent overrides are strictly prohibited.
- Any deviation from this specification MUST be documented, reviewed, and explicitly approved.

Bootnode Migration Policy

- Bootnodes may be designated as TEMPORARY or CANONICAL.
- Temporary bootnodes may be migrated, replaced, or decommissioned without protocol changes, client changes, or governance approval.
- Clients MUST treat bootnode identity, IP address, and location as mutable and MUST NOT encode assumptions about bootnode permanence.
- Bootnode retirement MAY occur without prior notice, provided that remaining bootnodes satisfy the minimum redundancy requirements defined in this document.

Change Accountability

- Any infrastructure change that is not reflected in this document is considered out of compliance and unsupported.
-

7. Additional Global Service Hostnames

Product	Global Hostname
Learning Hub / LMS	<i>https://learn.synergy-network.io</i>
uDEX	<i>https://udex.synergy-network.io</i>
Creator's Portal	<i>https://creators.synergy-network.io</i>
Governance / DAO Hub	<i>https://governance.synergy-network.io</i>
Global Chat / Realtime	<i>wss://realtime.synergy-network.io</i>

8. Global Router — Product Path Allocation

8.1 API Router

```
location /learn/      { proxy_pass http://learn_api; }
location /udex/      { proxy_pass http://udex_api; }
location /creators/  { proxy_pass http://creators_api; }
location /governance/ { proxy_pass http://governance_api; }
```

8.2 WebSocket Router

```
location /realtime/ {
    proxy_pass http://realtime_ws;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;
}
```

9. Global Nginx Upstream Placeholders

NOTE: Concrete port bindings are environment-specific and will be finalized in PART 3–5.

```
upstream learn_api      { server 127.0.0.1:0; }
upstream udex_api       { server 127.0.0.1:0; }
upstream creators_api    { server 127.0.0.1:0; }
upstream governance_api  { server 127.0.0.1:0; }
upstream realtime_ws     { server 127.0.0.1:0; }
```

10. Global WebSocket Policy

- All realtime services **MUST** route through *wss://ws.synergy-network.io*
 - Direct public WebSocket exposure is **prohibited**
 - WebSocket auth **MUST** be UMA-scoped
 - Cross-product chat streams **MUST** be namespace-isolated
-

SYNERGY DEVNET

11. Canonical DNS & Endpoint URLs

Surface	URL
Core RPC	https://devnet-core-rpc.synergy-network.io
Core WS	wss://devnet-core-ws.synergy-network.io
EVM RPC	https://devnet-evm-rpc.synergy-network.io
EVM WS	wss://devnet-evm-ws.synergy-network.io
REST API	https://devnet-api.synergy-network.io
Explorer UI	https://devnet-explorer.synergy-network.io
Explorer API	https://devnet-explorer-api.synergy-network.io
Indexer API	https://devnet-indexer.synergy-network.io
Faucet	https://devnet-faucet.synergy-network.io
Wallet API	https://devnet-wallet-api.synergy-network.io
SXCP API	https://devnet-sxcp-api.synergy-network.io
SXCP WS	wss://devnet-sxcp-ws.synergy-network.io
SynQ Verify	https://devnet-synq-verify.synergy-network.io

12. Canonical Port Allocation

12.1 L1 Node

Purpose	Port
P2P	38638
Core RPC	48638
Core WS	58638
Metrics	9090*

*localhost only

12.2 EVM RPC

Purpose	Port
HTTP	8545
WS	8546

12.3 Microservices

Service	Port
Portal API	3001
Faucet	3002
Wallet API	3003
Indexer Ingest	3010
Indexer API	3011
Explorer API	3020

SynQ Verify	3030
SXCP API	3040
SXCP WS	3041
Aegis Verify	3050
Aegis KMS	3051*

*PRIVATE, mTLS ONLY

13. Nginx — Devnet Upstream Map

```

upstream devnet_core_rpc      { server 127.0.0.1:48638; }
upstream devnet_core_ws      { server 127.0.0.1:58638; }

upstream devnet_evm_rpc      { server 127.0.0.1:8545; }
upstream devnet_evm_ws      { server 127.0.0.1:8546; }

upstream devnet_api          { server 127.0.0.1:3001; }
upstream devnet_explorer_ui  { server 127.0.0.1:80; }
upstream devnet_explorer_api { server 127.0.0.1:3020; }
upstream devnet_indexer_api  { server 127.0.0.1:3011; }
upstream devnet_wallet_api   { server 127.0.0.1:3003; }
upstream devnet_faucet       { server 127.0.0.1:3002; }

upstream devnet_sxcp_api     { server 127.0.0.1:3040; }
upstream devnet_sxcp_ws     { server 127.0.0.1:3041; }

upstream devnet_synq_verify  { server 127.0.0.1:3030; }

```

14. Nginx — Shared Proxy & WebSocket Headers

```

map $http_upgrade $connection_upgrade {
    default upgrade;
    ''      close;
}

proxy_set_header Host          $host;
proxy_set_header X-Real-IP     $remote_addr;
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
proxy_set_header X-Forwarded-Proto $scheme;
proxy_http_version 1.1;

```

15. Nginx — Devnet VHost Map (Exact DNS Match)

15.1 Core RPC / WS

```

server {
    listen 443 ssl http2;
    server_name devnet-core-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_core_rpc; }
}

server {

```

```

listen 443 ssl http2;
server_name devnet-core-ws.synergy-network.io;
include snippets/ssl.conf;
location / {
    proxy_pass http://devnet_core_ws;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;
}
}

```

15.2 EVM RPC / WS

```

server {
    listen 443 ssl http2;
    server_name devnet-evm-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_evm_rpc; }
}

server {
    listen 443 ssl http2;
    server_name devnet-evm-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://devnet_evm_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}

```

15.3 APIs, Explorer, Indexer

```

server {
    listen 443 ssl http2;
    server_name devnet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_api; }
}

server {
    listen 443 ssl http2;
    server_name devnet-explorer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_explorer_ui; }
}

server {
    listen 443 ssl http2;
    server_name devnet-explorer-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_explorer_api; }
}

server {
    listen 443 ssl http2;

```



```
server_name devnet-indexer.synergy-network.io;
include snippets/ssl.conf;
location / { proxy_pass http://devnet_indexer_api; }
}
```

15.4 Wallet, Faucet, SXCP, SynQ

```
server {
    listen 443 ssl http2;
    server_name devnet-wallet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_wallet_api; }
}

server {
    listen 443 ssl http2;
    server_name devnet-faucet.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_faucet; }
}

server {
    listen 443 ssl http2;
    server_name devnet-sxcp-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_sxcp_api; }
}

server {
    listen 443 ssl http2;
    server_name devnet-sxcp-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://devnet_sxcp_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}

server {
    listen 443 ssl http2;
    server_name devnet-synq-verify.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://devnet_synq_verify; }
}
```

SYNERGY TESTNET

16. Canonical DNS & Endpoint URLs

Surface	URL
Core RPC	https://testnet-core-rpc.synergy-network.io
Core WS	wss://testnet-core-ws.synergy-network.io
EVM RPC	https://testnet-evm-rpc.synergy-network.io
EVM WS	wss://testnet-evm-ws.synergy-network.io
REST API	https://testnet-api.synergy-network.io
Explorer UI	https://testnet-explorer.synergy-network.io
Explorer API	https://testnet-explorer-api.synergy-network.io
Indexer API	https://testnet-indexer.synergy-network.io
Faucet	https://testnet-faucet.synergy-network.io
Wallet API	https://testnet-wallet-api.synergy-network.io
SXCP API	https://testnet-sxcp-api.synergy-network.io
SXCP WS	wss://testnet-sxcp-ws.synergy-network.io
SynQ Verify	https://testnet-synq-verify.synergy-network.io

17. Canonical Port Allocation

17.1 L1 Node

Purpose	Port
P2P	38639
Core RPC	48639
Core WS	58639
Metrics	9091*

*localhost only

17.2 EVM RPC

Purpose	Port
HTTP	9545
WS	9546

17.3 Microservices

Service	Port
Portal API	3101
Faucet	3102
Wallet API	3103
Indexer Ingest	3110
Indexer API	3111
Explorer API	3120
SynQ Verify	3130
SXCP API	3140
SXCP WS	3141
Aegis Verify	3150
Aegis KMS	3151

*PRIVATE, mTLS ONLY

18. Nginx — Upstream Map

```
upstream testnet_core_rpc      { server 127.0.0.1:48639; }
upstream testnet_core_ws      { server 127.0.0.1:58639; }

upstream testnet_evm_rpc      { server 127.0.0.1:9545; }
upstream testnet_evm_ws      { server 127.0.0.1:9546; }

upstream testnet_api          { server 127.0.0.1:3101; }
upstream testnet_explorer_ui  { server 127.0.0.1:80; }
upstream testnet_explorer_api { server 127.0.0.1:3120; }
upstream testnet_indexer_api  { server 127.0.0.1:3111; }
upstream testnet_wallet_api   { server 127.0.0.1:3103; }
upstream testnet_faucet       { server 127.0.0.1:3102; }

upstream testnet_sxcp_api      { server 127.0.0.1:3140; }
upstream testnet_sxcp_ws      { server 127.0.0.1:3141; }

upstream testnet_synq_verify   { server 127.0.0.1:3130; }
```

19. Nginx — VHost Map (Exact DNS Match)

19.1 Core RPC / WS

```
server {
    listen 443 ssl http2;
    server_name testnet-core-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_core_rpc; }
}

server {
    listen 443 ssl http2;
    server_name testnet-core-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://testnet_core_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

19.2 EVM RPC / WS

```
server {
    listen 443 ssl http2;
    server_name testnet-evm-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_evm_rpc; }
}

server {
    listen 443 ssl http2;
```

```

server_name testnet-evm-ws.synergy-network.io;
include snippets/ssl.conf;
location / {
    proxy_pass http://testnet_evm_ws;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;
}
}

```

19.3 APIs, Explorer, Indexer

```

server {
    listen 443 ssl http2;
    server_name testnet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_api; }
}

server {
    listen 443 ssl http2;
    server_name testnet-explorer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_explorer_ui; }
}

server {
    listen 443 ssl http2;
    server_name testnet-explorer-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_explorer_api; }
}

server {
    listen 443 ssl http2;
    server_name testnet-indexer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_indexer_api; }
}

```

19.4 Wallet, Faucet, SXCP, SynQ

```

server {
    listen 443 ssl http2;
    server_name testnet-wallet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_wallet_api; }
}

server {
    listen 443 ssl http2;
    server_name testnet-faucet.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_faucet; }
}

```

```
server {
    listen 443 ssl http2;
    server_name testnet-sxcp-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_sxcp_api; }
}

server {
    listen 443 ssl http2;
    server_name testnet-sxcp-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://testnet_sxcp_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}

server {
    listen 443 ssl http2;
    server_name testnet-synq-verify.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://testnet_synq_verify; }
}
```

SYNERGY MAINNET-BETA

20. Canonical DNS & Endpoint URLs

Surface	URL
Core RPC	https://beta-core-rpc.synergy-network.io
Core WS	wss://beta-core-ws.synergy-network.io
EVM RPC	https://beta-evm-rpc.synergy-network.io
EVM WS	wss://beta-evm-ws.synergy-network.io
REST API	https://beta-api.synergy-network.io
Explorer UI	https://beta-explorer.synergy-network.io
Explorer API	https://beta-explorer-api.synergy-network.io
Indexer API	https://beta-indexer.synergy-network.io
Wallet API	https://beta-wallet-api.synergy-network.io
SXCP API	https://beta-sxcp-api.synergy-network.io
SXCP WS	wss://beta-sxcp-ws.synergy-network.io
SynQ Verify	https://beta-synq-verify.synergy-network.io

19. Canonical Port Allocation

19.1 L1 Node

Purpose	Port
P2P	38640
Core RPC	48640
Core WS	58640
Metrics	9092*

**localhost only*

19.2 EVM RPC

Purpose	Port
HTTP	10545
WS	10546

19.3 Microservices

Service	Port
Portal API	3201
Wallet API	3203
Indexer Ingest	3210
Indexer API	3211
Explorer API	3220
SynQ Verify	3230
SXCP API	3240
SXCP WS	3241
Aegis Verify	3250
Aegis KMS	3251

**PRIVATE, mTLS ONLY*

20. Nginx — Upstream Map

```
upstream beta_core_rpc      { server 127.0.0.1:48640; }
upstream beta_core_ws      { server 127.0.0.1:58640; }

upstream beta_evm_rpc      { server 127.0.0.1:10545; }
upstream beta_evm_ws      { server 127.0.0.1:10546; }

upstream beta_api          { server 127.0.0.1:3201; }
upstream beta_explorer_ui  { server 127.0.0.1:80; }
upstream beta_explorer_api { server 127.0.0.1:3220; }
upstream beta_indexer_api  { server 127.0.0.1:3211; }
upstream beta_wallet_api   { server 127.0.0.1:3203; }

upstream beta_sxcp_api     { server 127.0.0.1:3240; }
upstream beta_sxcp_ws     { server 127.0.0.1:3241; }

upstream beta_synq_verify  { server 127.0.0.1:3230; }
```

21. Nginx — VHost Map (Exact DNS Match)

21.1 Core RPC / WS

```
server {
    listen 443 ssl http2;
    server_name beta-core-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_core_rpc; }
}

server {
    listen 443 ssl http2;
    server_name beta-core-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://beta_core_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

21.2 EVM RPC / WS

```
server {
    listen 443 ssl http2;
    server_name beta-evm-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_evm_rpc; }
}

server {
    listen 443 ssl http2;
    server_name beta-evm-ws.synergy-network.io;
```

```

    include snippets/ssl.conf;
    location / {
        proxy_pass http://beta_evm_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}

```

21.3 APIs, Explorer, Indexer

```

server {
    listen 443 ssl http2;
    server_name beta-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_api; }
}

server {
    listen 443 ssl http2;
    server_name beta-explorer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_explorer_ui; }
}

server {
    listen 443 ssl http2;
    server_name beta-explorer-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_explorer_api; }
}

server {
    listen 443 ssl http2;
    server_name beta-indexer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_indexer_api; }
}

```

21.4 Wallet, SXCP, SynQ

```

server {
    listen 443 ssl http2;
    server_name beta-wallet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_wallet_api; }
}

server {
    listen 443 ssl http2;
    server_name beta-sxcp-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_sxcp_api; }
}

server {

```



```
listen 443 ssl http2;
server_name beta-sxcp-ws.synergy-network.io;
include snippets/ssl.conf;
location / {
    proxy_pass http://beta_sxcp_ws;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;
}

server {
    listen 443 ssl http2;
    server_name beta-synq-verify.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://beta_synq_verify; }
}
```

SYNERGY MAINNET

22. Canonical DNS & Endpoint URLs

Surface	URL
Core RPC	https://core-rpc.synergy-network.io
Core WS	wss://core-ws.synergy-network.io
EVM RPC	https://evm-rpc.synergy-network.io
EVM WS	wss://evm-ws.synergy-network.io
REST API	https://api-mainnet.synergy-network.io
Explorer UI	https://explorer.synergy-network.io
Explorer API	https://explorer-api.synergy-network.io
Indexer API	https://indexer.synergy-network.io
Wallet API	https://wallet-api.synergy-network.io
SXCP API	https://sxcapi.synergy-network.io
SXCP WS	wss://sxcapi-ws.synergy-network.io
SynQ Verify	https://synq-verify.synergy-network.io
Learning Hub	https://learn.synergy-network.io
uDEX	https://udex.synergy-network.io
Creator's Portal	https://creators.synergy-network.io
Governance Hub	https://governance.synergy-network.io
Realtime / Chat	wss://realtime.synergy-network.io

23. Canonical Port Allocation

23.1 L1 Node

Purpose	Port
P2P	38641
Core RPC	48641
Core WS	58641
Metrics	9093*

*localhost only

23.2 EVM RPC

Purpose	Port
HTTP	11545
WS	11546

23.3 Microservices

Service	Port
Portal API	3301
Wallet API	3303
Indexer Ingest	3310
Indexer API	3311
Explorer API	3320
SynQ Verify	3330
SXCP API	3340
SXCP WS	3341
Learning Hub API	3350
uDEX API	3360
Creator's API	3370
Governance API	3380
Realtime WS	3390
Aegis Verify	3400
Aegis KMS	3401*

*PRIVATE, mTLS ONLY

24. Nginx — Upstream Map

```
upstream mainnet_core_rpc      { server 127.0.0.1:48641; }
upstream mainnet_core_ws      { server 127.0.0.1:58641; }

upstream mainnet_evm_rpc      { server 127.0.0.1:11545; }
upstream mainnet_evm_ws      { server 127.0.0.1:11546; }

upstream mainnet_api          { server 127.0.0.1:3301; }
upstream mainnet_explorer_ui  { server 127.0.0.1:80; }
upstream mainnet_explorer_api { server 127.0.0.1:3320; }
upstream mainnet_indexer_api  { server 127.0.0.1:3311; }
upstream mainnet_wallet_api   { server 127.0.0.1:3303; }

upstream mainnet_sxcp_api     { server 127.0.0.1:3340; }
upstream mainnet_sxcp_ws     { server 127.0.0.1:3341; }

upstream mainnet_synq_verify  { server 127.0.0.1:3330; }

upstream learn_api            { server 127.0.0.1:3350; }
upstream udex_api             { server 127.0.0.1:3360; }
upstream creators_api         { server 127.0.0.1:3370; }
upstream governance_api       { server 127.0.0.1:3380; }
upstream realtime_ws         { server 127.0.0.1:3390; }
```

25. Nginx — VHost Map (Exact DNS Match)

25.1 Core RPC / WS

```
server {
    listen 443 ssl http2;
    server_name core-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_core_rpc; }
}

server {
    listen 443 ssl http2;
    server_name core-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://mainnet_core_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

25.2 EVM RPC / WS

```
server {
    listen 443 ssl http2;
    server_name evm-rpc.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_evm_rpc; }
}

server {
    listen 443 ssl http2;
    server_name evm-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://mainnet_evm_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

25.3 APIs, Explorer, Indexer

```
server {
    listen 443 ssl http2;
    server_name api-mainnet.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_api; }
}

server {
    listen 443 ssl http2;
    server_name explorer.synergy-network.io;
    include snippets/ssl.conf;
```

```

    location / { proxy_pass http://mainnet_explorer_ui; }
}

server {
    listen 443 ssl http2;
    server_name explorer-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_explorer_api; }
}

server {
    listen 443 ssl http2;
    server_name indexer.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_indexer_api; }
}

```

25.4 Wallet, SXCP, SynQ

```

server {
    listen 443 ssl http2;
    server_name wallet-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_wallet_api; }
}

server {
    listen 443 ssl http2;
    server_name sxcp-api.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_sxcp_api; }
}

server {
    listen 443 ssl http2;
    server_name sxcp-ws.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://mainnet_sxcp_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}

server {
    listen 443 ssl http2;
    server_name synq-verify.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://mainnet_synq_verify; }
}

```

25.5 Learning Hub, uDEX, Creator's Portal, Governance, Realtime

```
server {
    listen 443 ssl http2;
    server_name learn.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://learn_api; }
}

server {
    listen 443 ssl http2;
    server_name udex.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://udex_api; }
}

server {
    listen 443 ssl http2;
    server_name creators.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://creators_api; }
}

server {
    listen 443 ssl http2;
    server_name governance.synergy-network.io;
    include snippets/ssl.conf;
    location / { proxy_pass http://governance_api; }
}

server {
    listen 443 ssl http2;
    server_name realtime.synergy-network.io;
    include snippets/ssl.conf;
    location / {
        proxy_pass http://realtime_ws;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
    }
}
```

DNS Records

Global

- *api.synergy-network.io*
- *rpc.synergy-network.io*
- *ws.synergy-network.io*
- *explorer.synergy-network.io*
- *indexer.synergy-network.io*
- *assets.synergy-network.io*
- *docs.synergy-network.io*
- *validators.synergy-network.io*
- *status.synergy-network.io*
- *realtime.synergy-network.io*

Devnet

- *devnet-core-rpc.synergy-network.io*
- *devnet-core-ws.synergy-network.io*
- *devnet-evm-rpc.synergy-network.io*
- *devnet-evm-ws.synergy-network.io*
- *devnet-api.synergy-network.io*
- *devnet-explorer.synergy-network.io*
- *devnet-explorer-api.synergy-network.io*
- *devnet-indexer.synergy-network.io*
- *devnet-wallet-api.synergy-network.io*
- *devnet-faucet.synergy-network.io*
- *devnet-sxcp-api.synergy-network.io*
- *devnet-sxcp-ws.synergy-network.io*
- *devnet-synq-verify.synergy-network.io*

Testnet

- *testnet-core-rpc.synergy-network.io*
- *testnet-core-ws.synergy-network.io*
- *testnet-evm-rpc.synergy-network.io*
- *testnet-evm-ws.synergy-network.io*
- *testnet-api.synergy-network.io*
- *testnet-explorer.synergy-network.io*
- *testnet-explorer-api.synergy-network.io*
- *testnet-indexer.synergy-network.io*
- *testnet-wallet-api.synergy-network.io*
- *testnet-faucet.synergy-network.io*
- *testnet-sxcp-api.synergy-network.io*
- *testnet-sxcp-ws.synergy-network.io*
- *testnet-synq-verify.synergy-network.io*

Mainnet-Beta

- *beta-core-rpc.synergy-network.io*
- *beta-core-ws.synergy-network.io*
- *beta-evm-rpc.synergy-network.io*
- *beta-evm-ws.synergy-network.io*
- *beta-api.synergy-network.io*
- *beta-explorer.synergy-network.io*
- *beta-explorer-api.synergy-network.io*
- *beta-indexer.synergy-network.io*
- *beta-wallet-api.synergy-network.io*
- *beta-sxcp-api.synergy-network.io*
- *beta-sxcp-ws.synergy-network.io*
- *beta-synq-verify.synergy-network.io*

Mainnet

- *core-rpc.synergy-network.io*
- *core-ws.synergy-network.io*
- *evm-rpc.synergy-network.io*
- *evm-ws.synergy-network.io*
- *api-mainnet.synergy-network.io*
- *explorer.synergy-network.io*
- *explorer-api.synergy-network.io*
- *indexer.synergy-network.io*
- *wallet-api.synergy-network.io*
- *sxcp-api.synergy-network.io*
- *sxcp-ws.synergy-network.io*
- *synq-verify.synergy-network.io*
- *learn.synergy-network.io*
- *udex.synergy-network.io*
- *creators.synergy-network.io*
- *governance.synergy-network.io*
- *realtime.synergy-network.io*

APPENDIX A — LEAN 15 DEVNET AUTHORITATIVE MACHINE-BY-MACHINE CONFIGURATION MATRIX

Authority: This appendix is mandatory and supersedes generic devnet examples. All Lean-15 deployments MUST match this matrix exactly unless this manual is explicitly revised.

A.1 Global Lean-15 Configuration Constants

- network.id = 338638; blockchain.chain_id = 338638; network.name = synergy-devnet
- block_time = 5 seconds; consensus.algorithm = Proof of Synergy
- max_peers = 100; heartbeat_interval = 30
- consensus.epoch_length = 30000; validator_cluster_size = 7; max_validators = 15
- synergy_score_decay_rate = 0.05; vrf_seed_epoch_interval = 1000
- max_synergy_points_per_epoch = 100; max_tasks_per_validator = 10
- reward weighting = task_accuracy 0.5, uptime 0.3, collaboration 0.2
- rpc.enable_http = true; rpc.enable_ws = true; rpc.enable_grpc = true
- storage.pruning_interval = 86400; cors_origins = [*]
- Bootnode policy: machine-01 has no bootnodes; machine-02 uses machine-01; machines 03-15 use machine-01 + machine-02

A.2 Authoritative Machine-by-Machine Configuration Matrix

Machine	Node ID	Role	Type	Host	Ports P2P/RPC/WS/gRPC/Disc	AutoReg	Prune	VRF	Bootnodes
machine-01	consensus-01	validator	validator	machine01.synergy-devnet.local	38638/48638/58638/50051/39638	true	false	true	[]
machine-02	consensus-02	validator	validator	machine02.synergy-devnet.local	38639/48639/58639/50052/39639	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638]
machine-03	consensus-03	validator	validator	machine03.synergy-devnet.local	38640/48640/58640/50053/39640	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639]
machine-04	consensus-04	validator	validator	machine04.synergy-devnet.local	38641/48641/58641/50054/39641	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640]
machine-05	consensus-05	validator	validator	machine05.synergy-devnet.local	38642/48642/58642/50055/39642	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641]
machine-06	interop-01	relayer	relayer	machine06.synergy-devnet.local	38643/48643/58643/50056/39643	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642]
machine-07	interop-02	cross-chain-verifier	cross-chain-verifier	machine07.synergy-devnet.local	38644/48644/58644/50057/39644	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642, snr://bootstrap@machine06.synergy-devnet.local:38643]
machine-08	interop-03	oracle	oracle	machine08.synergy-devnet.local	38645/48645/58645/50058/39645	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642, snr://bootstrap@machine06.synergy-devnet.local:38643, snr://bootstrap@machine07.synergy-devnet.local:38644]
machine-09	interop-04	witness	witness	machine09.synergy-devnet.local	38646/48646/58646/50059/39646	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642, snr://bootstrap@machine06.synergy-devnet.local:38643, snr://bootstrap@machine07.synergy-devnet.local:38644, snr://bootstrap@machine08.synergy-devnet.local:38645]
machine-10	governance-01	committee	committee	machine10.synergy-devnet.local	38647/48647/58647/50060/39647	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642, snr://bootstrap@machine06.synergy-devnet.local:38643, snr://bootstrap@machine07.synergy-devnet.local:38644, snr://bootstrap@machine08.synergy-devnet.local:38645, snr://bootstrap@machine09.synergy-devnet.local:38646]
machine-11	governance-02	security-council	security-council	machine11.synergy-devnet.local	38648/48648/58648/50061/39648	true	false	true	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://bootstrap@machine02.synergy-devnet.local:38639, snr://bootstrap@machine03.synergy-devnet.local:38640, snr://bootstrap@machine04.synergy-devnet.local:38641, snr://bootstrap@machine05.synergy-devnet.local:38642, snr://bootstrap@machine06.synergy-devnet.local:38643, snr://bootstrap@machine07.synergy-devnet.local:38644, snr://bootstrap@machine08.synergy-devnet.local:38645, snr://bootstrap@machine09.synergy-devnet.local:38646, snr://bootstrap@machine10.synergy-devnet.local:38647]

Machine	Node ID	Role	Type	Host	Ports P2P/RPC/WS/gRPC/Disc	AutoReg	Prune	VRF	Bootnodes
machine-12	pqc-01	pqc-crypto	pqc-crypto	machine12.synergy-devnet.local	38649/48649/58649/50062/39649	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://l
machine-13	service-01	rpc-gateway	rpc-gateway	machine13.synergy-devnet.local	38650/48650/58650/50063/39650	false	true	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://l
machine-14	service-02	indexer	indexer	machine14.synergy-devnet.local	38651/48651/58651/50064/39651	false	false	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://l
machine-15	service-03	observer	observer	machine15.synergy-devnet.local	38652/48652/58652/50065/39652	false	false	false	[snr://bootstrap@machine01.synergy-devnet.local:38638, snr://l

A.3 Firewall and Exposure Requirements (Mandatory)

- Open inter-node TCP ports for all machines: 38638-38652 (P2P), 39638-39652 (discovery), 50051-50065 (gRPC as required).
- Expose RPC/WS (48638-48652 / 58638-58652) only according to environment security policy; public exposure should be limited to service surfaces behind nginx/TLS.
- Metrics endpoints must remain localhost-only and are not part of public ingress.
- If DNS records are used, each hostname must resolve from every peer and from monitoring infrastructure exactly as listed in the matrix.